

Parashar21 Pipeline

Comprehensive Alphabetical Function Reference

1. Documentation Scope and Architectural Framework

This reference guide provides the definitive structural map of the Parashar21 astrological pipeline. As a system designed to transform raw birth data—Date, Time, and Place of Birth—into high-value outputs for human-readable reports and Large Language Model (LLM) prompts, its strategic integrity relies on a modular, decoupled architecture. By segregating the **Astronomical Backend** ([p21swe](#)), the **Interpretive Logic Tier** ([p21utils](#)), and the **Presentation Tier** ([p21utilsR](#), [p21LLM](#)), the pipeline ensures that precision-critical calculations remain isolated from formatting logic.

The documentation methodology emphasizes technical transparency through rigorous caller/callee tracing. This allows maintainers to track how the **Global State Container** ([p21.py](#)) is initialized, modified, and read across the lifecycle of a chart. Central to this process is the Python-Swiss Ephemeris bridge, which translates tropical coordinates into the sidereal framework required for Vedic (Jyotisha) analysis. The following reference detail identifies how these modules interact to establish a single "Source of Truth" within the pipeline's data persistence layer.

2. Function Reference: A - C

The "A through C" range encompasses the critical initialization and data-structuring phases. These functions are responsible for the first-mile transformation of birth data into an enriched state, populating the global dictionaries that define planetary dignity and house lordship.

- **AddBhavaAspects** ([p21LLM.py](#))
 - **Caller:** [R601_GenerateLLMInput05](#)
 - **Parameters:** [lines](#) (List): The string buffer for LLM input.
 - **Implicit:** Global State ([p21.BAspectedBy2](#))
 - **Description:** Appends descriptions of which planets are aspecting specific houses (Bhavas) to the LLM text stream.
- **AddBhavaAspectedByLords** ([p21LLM.py](#))
 - **Caller:** [R601_GenerateLLMInput05](#)
 - **Parameters:** [lines](#) (List): The string buffer.
 - **Implicit:** Global State ([p21.BAspectedByBL2](#))

- **Description:** Identifies and documents which house lords are specifically aspecting which Bhavas.
- **AddDashaFull** (p21LLM.py)
 - **Caller:** R602_GenerateLLMInput01
 - **Parameters:** `lines` (List): The string buffer.
 - **Implicit:** Global State (p21.VimDasha)
 - **Description:** Formats the complete nested Vimshottari Dasha/Antardasha sequence into a structured text format for AI ingestion.
- **AddHeader** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05, R602_GenerateLLMInput01
 - **Parameters:** `lines` (List), `analysisMode` (String), `title` (String).
 - **Description:** Generates the LLM file header, including metadata like subject name and gender retrieved via `GetGender`.
- **AddHouseLordRelationships** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** `lines` (List).
 - **Implicit:** Global State (p21.GrahaLordBhav)
 - **Description:** Documents the relationships between specific house lords and the planets they interact with.
- **AddHouseLordship** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** `lines` (List).
 - **Implicit:** Global State (p21.GrahaLordBhav)
 - **Description:** Explicitly lists planetary ownership of houses for the current chart.
- **AddLordLordRelationships** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** `lines` (List).
 - **Implicit:** Global State (p21.BLConjunctsBL2)
 - **Description:** Captures and formats interactions and conjunctions between different house lords.
- **AddPlanetaryAspects** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** `lines` (List).
 - **Implicit:** Global State (p21.GAspects2)
 - **Description:** Appends data regarding which planets are casting aspects onto other specific planets.
- **AddPlanetaryConjunctions** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** `lines` (List).
 - **Implicit:** Global State (p21.GConjunctsG2)

- **Description:** Identifies and formats groups of planets residing in the same Rashi.
- **AddPlanetaryInformation** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** lines (List).
 - **Implicit:** Global State (p21.GRashiN, p21.GLon, p21.GRet)
 - **Description:** Provides foundational planetary data including Rashi placement, longitude, and retrograde status.
- **AddPlanetsAspectedBy** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** lines (List).
 - **Implicit:** Global State (p21.GAspectedBy2)
 - **Description:** Documents the inverse aspect relationship—identifying which planets are being aspected by which others.
- **AddYogas** (p21LLM.py)
 - **Caller:** R601_GenerateLLMInput05
 - **Parameters:** lines (List).
 - **Implicit:** Global State (p21.yogsFound)
 - **Description:** Lists the specific astrological Yogas identified in the chart via logic from p21YogInfo.
- **addToD** (p21utils.py)
 - **Parameters:** x (Key), D (Dictionary), y (Value).
 - **Description:** Internal logic helper that manages sets within dictionaries, ensuring keys are initialized before values are appended.
- **appendDict** (p21utils.py)
 - **Parameters:** d1 (Target), d2 (Source).
 - **Description:** Merges the contents of two dictionaries, used for aggregating logic parameters.
- **C01_configSWE** (p21swe.py)
 - **Description:** Configures the Swiss Ephemeris library and initializes global flags for astronomical calculation.
- **C02_parsePersonData** (p21swe.py)
 - **Caller:** C61_Cast2JSON, setTransitData
 - **Parameters:** P (Pandas Series): A single row of birth data.
 - **Description:** Extracts coordinates and time components from raw input into the pipeline's operational variables.
- **C03_convertDates** (p21swe.py)
 - **Caller:** C61_Cast2JSON, setTransitData
 - **Implicit:** Updates p21.natalUT, p21.ayanamsha
 - **Description:** Converts local time to UTC/Julian Date and computes the Lahiri Ayanamsha for sidereal adjustment.
- **C04_calculateGrahaPositions** (p21swe.py)

- **Caller:** C61_Cast2JSON, setTransitData
 - **Implicit:** Updates p21.bLon, p21.bRet
 - **Description:** Computes raw longitudes and retrograde status for planets (including Rahu as a Mean Node).
- **C05_buildGLonGRet** (p21swe.py)
 - **Caller:** setTransitData
 - **Callees:** p21Sub
 - **Description:** Constructs the sidereal longitude dictionary. **Technical Note:** Currently uses Ayanamsha subtraction; migration to SEFLG_SIDEREAL is the preferred long-term standard.
- **C10_DetermineBhavs** (p21utils.py)
 - **Caller:** C61_Cast2JSON
 - **Implicit:** Updates p21.BhavN, p21.BhavA
 - **Description:** Assigns Rashis to the twelve houses (Bhavas), establishing the house framework.
- **C11_DetermineLord** (p21utils.py)
 - **Caller:** C61_Cast2JSON
 - **Implicit:** Updates p21.Lord, p21.LordRashiN
 - **Description:** Identifies the ruling planet (Lord) for every house based on the assigned Rashi.
- **C12_BhavOfGraha_Lord** (p21utils.py)
 - **Caller:** C61_Cast2JSON
 - **Implicit:** Global State (p21.LordOf)
 - **Description:** Maps planets to the Bhavas they rule; handles dual lordship for planets other than the Sun and Moon.
- **C12A_StoreRashiOfGraha** (p21utils.py)
 - **Caller:** C61_Cast2JSON
 - **Description:** Finalizes the storage of planetary Rashi and Bhava residence.
- **C21_DeterminePositions** (p21utils.py)
 - **Caller:** C61_Cast2JSON
 - **Callees:** C21A through C21G
 - **Description:** Orchestrates the complex evaluation of planetary dignities (Exaltation, Debilitation, Moolatrikona).
- **C21A_checkGexa** (p21utils.py)
 - **Parameters:** x (Planet), level (Default: 'low')
 - **Implicit:** Global State (p21.exaR)
 - **Description:** Checks if a planet is in its Rashi of exaltation using the exaR dictionary.
- **C21B_checkLexa** (p21utils.py)
 - **Parameters:** x (Planet Lord ID)

- **Implicit:** Global State (`p21.exaltG`)
 - **Description:** Evaluates if a house Lord is currently in an exalted state.
- **C21C_checkGdeb** (`p21utils.py`)
 - **Implicit:** Global State (`p21.debR`)
 - **Description:** Checks if a planet is in its Rashi of debilitation using the `debR` dictionary.
- **C21D_checkLdeb** (`p21utils.py`)
 - **Implicit:** Global State (`p21.debilG`)
 - **Description:** Evaluates if a house Lord is currently in a debilitated state.
- **C21E_checkm3G** (`p21utils.py`)
 - **Implicit:** Global State (`p21.mool3R`)
 - **Description:** Checks if a planet is in its Moolatrikona Rashi based on the `mool3R` mapping.
- **C21F_checkOwnHG** (`p21utils.py`)
 - **Implicit:** Global State (`p21.LordOf`)
 - **Description:** Determines if a planet is residing in its own Rashi (Swakshetra).
- **C21G_checkfen** (`p21utils.py`)
 - **Parameters:** `x` (Planet), `Z` (Relationship Dict)
 - **Description:** Categorizes the relationship (Friend, Enemy, Neutral) of a planet with its Rashi Lord.
- **C31_DetermineAspects** (`p21utils.py`)
 - **Caller:** `C61_Cast2JSON`
 - **Description:** Calculates the angular relationships (aspects) between planets and houses.
- **C41_DetermineConjuncts** (`p21utils.py`)
 - **Caller:** `C61_Cast2JSON`
 - **Description:** Identifies groupings of planets and lords occupying the same Rashi.
- **C51_DetermineBenMal** (`p21utils.py`)
 - **Caller:** `C61_Cast2JSON`
 - **Description:** Classifies planets as Benefics or Malefics based on the specific Lagna (Ascendant).
- **C61_Cast2JSON** (`p21swe.py`)
 - **Caller:** Orchestration Layer
 - **Parameters:** `input_df` (Pandas DataFrame)
 - **Description:** The primary state-initializer. It converts raw CSV data into an enriched JSON artifact, serving as the pivot point for all downstream analysis.
- **csidtil** (`p21utils.py`)
 - **Description:** Utility to convert sets within dictionaries to lists, ensuring JSON/MongoDB serialization compatibility.

3. Function Reference: G - L

This section marks the shift from raw data calculation to logical synthesis. In this "So What?" layer, data is transformed into information—converting coordinates into Nakshatras and Dashas, and acting as the gateway to harmonic analysis like Navamsha (9th divisional chart).

- **GenAshtakVargaData_v1** ([p21utils.py](#))
 - **Description:** Deprecated version of the Ashtakvarga point calculation.
- **genChart** (Orchestration: [p21_51_chart2llm.txt](#))
 - **Callees:** [R12B](#) (Drawing), [R01_CreateReportDoc](#)
 - **Parameters:** [ChartStyle](#) (String)
 - **Description:** High-level notebook orchestrator that initiates both visual rendering and document generation.
- **generateChartTxt** ([p21utilsR.py](#))
 - **Caller:** [R12B_drawChart](#) variants
 - **Description:** Populates the 12 Rashi boxes with planetary abbreviations and retrograde markers (/R) for visual display.
- **GetAnalysisMode** ([p21LLM.py](#))
 - **Description:** Distinguishes between "Natal Analysis" and "Gochar Analysis" by comparing [pName](#) and [gName](#).
- **getChartData** (Orchestration: [p21_51_chart2llm.txt](#))
 - **Description:** Loads the persisted [peopleData.json](#) artifact into memory.
- **GetDasha** ([p21utils.py](#))
 - **Implicit:** [p21.SubMoonLong](#), [p21.NksGap](#) (13.33°)
 - **Description:** Converts Moon longitude into temporal periods. It calculates [DashaConsumedDeg](#) to determine exact start/end dates for Vimshottari periods.
- **GetGender** ([p21LLM.py](#))
 - **Description:** Maps gender codes to human-readable strings for report headers.
- **getTransitData** ([p21utils.py](#))
 - **Parameters:** [cDT](#) (Datetime)
 - **Description:** Sets up parameters for Gochar analysis, defaulting to current time/coordinates.
- **GList** ([p21LLM.py](#))
 - **Description:** Translates a list of planet codes into full English names for LLM clarity.
- **GName** ([p21LLM.py](#))
 - **Description:** Translates a single planet code (e.g., "Ju" to "Jupiter").
- **Long2Rashi** ([p21utils.py](#))
 - **Description:** Converts 0-360° longitude into Rashi indices; critical for Navamsha analysis by dividing the zodiac into 108 partitions (3.33° each).

4. Function Reference: N - R

The functions in the "N through R" block manage the **Presentation Tier**. They transform the internal state (persisted as JSON) into end-user artifacts, specifically the Microsoft Word reports and formatted LLM input files.

- **NextDasha** ([p21utils.py](#))
 - **Description:** Returns the next planet in the Vimshottari sequence ([DashaSeq](#)).
- **p21Sub** ([p21swe.py](#))
 - **Parameters:** *x*, *y* (Angles)
 - **Description:** Subtracts angles and normalizes the result to the 0-360° range; used for sidereal adjustments.
- **R01_CreateReportDoc** ([p21utilsR.py](#))
 - **Callees:** [python-docx](#) functions, [R512_FormatPage](#)
 - **Description:** The primary controller for the Microsoft Word output, handling document object initialization and layout.
- **R11_LocateGrahInRashi** ([p21utils.py](#))
 - **Caller:** [setTransitData](#), [C61_Cast2JSON](#)
 - **Description:** Updates the global state with the final Rashi placements for all planets.
- **R12B_drawChart_Bengal / _North / _South** ([p21utilsR.py](#))
 - **Description:** Renders the horoscope in regional visual formats (Eastern Indian, North Indian Diamond, or South Indian Square).
- **R13A_ShowTrueDict** ([p21utilsR.py](#))
 - **Description:** Filter utility that returns only the "True" keys from a dignity dictionary.
- **R13B_ListPositions / R13C_ListPositions** ([p21utilsR.py](#))
 - **Description:** Formatting utilities for planetary lists in human-readable documents.
- **R13D_ListContents** ([p21utilsR.py](#))
 - **Description:** Converts list data into tab-separated strings for tabular report sections.
- **R30_LocateYogs** ([p21utilsR.py](#))
 - **Implicit:** [p21YogInfo](#) data
 - **Description:** Scans the chart against a library of classical Yoga conditions (e.g., Raman's "300 Important Combinations").
- **R511_parseChartData** ([p21utilsR.py](#))
 - **Description:** Extracts JSON data into specific variables prepared for report paragraph insertion.
- **R512_FormatPage / _2A / _2B / _3** ([p21utilsR.py](#))
 - **Description:** A family of layout functions that manage the specific content of different pages in the Word report (Natal data, Dashas, Dignities).

- **R601_GenerateLLMInput05** ([p21LLM.py](#))
 - **Description:** The primary AI-bridge; aggregates all planetary, house, and Yoga logic into a single structured text file for LLM interpretation.
- **R602_GenerateLLMInput01** ([p21LLM.py](#))
 - **Description:** A specialized LLM-bridge focusing exclusively on temporal Dasha data.
- **RashiGapA** ([p21utils.py](#))
 - **Description:** Calculates the sign distance (1-12) between two Rashis for aspect logic.
- **RashiN2A** ([p21.py](#))
 - **Description:** Converts a Rashi index to its Sanskrit-derived name (e.g., 1 to "Mesh").
- **RName** ([p21LLM.py](#))
 - **Description:** Converts a Rashi index to its English name (e.g., 1 to "Aries").

5. Function Reference: S - Z

These terminal functions focus on specialized data structures and diagnostic tracing required for Gochar (Transit) analysis.

- **SarvaAshtakVarga** ([p21utils.py](#))
 - **Description:** Calculates the cumulative Ashtakvarga points for every Rashi.
Validation Gate: Every chart must aggregate to a checksum value of 337.
- **setTransitData** ([p21utils.py](#))
 - **Caller:** Gochar Orchestration
 - **Description:** Re-initializes the global state for a transit timestamp. Notably, it sets the Lagna (Ascendant) to the subject's natal Moon position, as per standard Gochar methodology.
- **tracer** ([p21utilsR.py](#))
 - **Description:** Diagnostic utility for verifying [ChartType](#) and system flow during execution.

6. Global Data Structures and Module Dependency Matrix

The pipeline's "Shared Vocabulary" is maintained through global variables defined in [p21.py](#). Constants such as [LordOf](#) (Rashi ownership) and [Graha](#) (Celestial bodies) ensure consistency as data moves from the astronomical backend to the presentation tier.

Module Dependency Matrix

Module	Primary Responsibility	State Interaction
p21.py	Global State Container	Initializer: Defines the shared vocabulary.
p21swe.py	Astronomical Backend	Writer: Interacts with Swiss Ephemeris to set positions.
p21utils.py	Interpretive Logic Engine	Read/Write: Calculates dignities/dashas; enriches state.
p21utilsR.py	Presentation (Visual/Word)	Read-Only: Formats state for human artifacts.
p21LLM.py	AI Presentation Tier	Read-Only: Formats state for Large Language Models.
Orchestration Layer	Pipeline Controller	Orchestrator: Executes the sequence from CSV to Artifact.

Architectural Robustness

The Parashar21 pipeline achieves robustness through the strict separation of **Calculation** (swe), **Logic** (utils), and **Presentation** (utilsR/LLM). By persisting the "Source of Truth" in `peopleData.json`, the system allows the same verified data to be simultaneously dispatched to a Word document and an LLM without redundant calculation. This modularity ensures that future updates to astronomical standards (e.g., sidereal flag adjustments) or AI formatting requirements can be implemented without destabilizing the core astrological logic.

Exhaustive Function Call Hierarchy Report

Primary Driver Execution: p21_51_chart2llm.txt

- `p21_51_chart2llm.txt`
 - `p21swe.C01_configSWE()`

Natal Chart Generation Pass

- `p21swe.C61_Cast2JSON(input_df)`
 - `p21swe.C02_parsePersonData(personData)`
 - `p21swe.C03_convertDates()`
 - `p21swe.C04_calculateGrahaPositions()`
 - `p21swe.C05_buildGLonGRet()`
- `p21utils.GetDasha()`
- `p21_51_chart2llm.getChartData()`
- `p21_51_chart2llm.genChart(ChartStyle)`
 - `p21utilsR.R12B_drawChart_Bengal()`
 - `p21utilsR.R30_LocateYogs()`
 - `p21utilsR.R01_CreateReportDoc(cqs, pS, repStyle)`

Gochar (Transit) Chart Generation Pass

- `p21utils.setTransitData(c)`
 - `p21swe.C02_parsePersonData(c)`
 - `p21swe.C03_convertDates()`
 - `p21swe.C04_calculateGrahaPositions()`
 - `p21swe.C05_buildGLonGRet()`
 - `p21utils.R11_LocateGrahaInRashi()`
- `p21swe.C61_Cast2JSON(df)`
- `p21_51_chart2llm.genChart(ChartStyle)`

Downstream LLM Input Generation (p21LLM Module)

- `p21LLM.R601_GenerateLLMInput05(filename)`
 - `p21LLM.AddHeader(lines, analysisMode, title)`

- **p21LLM.AddHouseLordship**(lines)
- **p21LLM.AddPlanetaryInformation**(lines)
- **p21LLM.AddPlanetaryAspects**(lines)
- **p21LLM.AddPlanetsAspectedBy**(lines)
- **p21LLM.AddPlanetaryConjunctions**(lines)
- **p21LLM.AddHouseLordRelationships**(lines)
- **p21LLM.AddLordLordRelationships**(lines)
- **p21LLM.AddBhavaAspects**(lines)
- **p21LLM.AddBhavaAspectedByLords**(lines)
- **p21LLM.AddYogas**(lines)
- **p21LLM.AddDashaFull**(lines)
- **p21LLM.R602_GenerateLLMInput01**(filename)

Utility and Structural Support Functions

- **p21swe Module**
 - **p21swe.p21Sub**(x, y)
- **p21utils Module**
 - **p21utils.C10_DetermineBhavs**()
 - **p21utils.C11_DetermineLord**()
 - **p21utils.C21_DeterminePositions**()
 - **p21utils.C31_DetermineAspects**()
 - **p21utils.C41_DetermineConjuncts**()
 - **p21utils.C51_DetermineBenMal**(